

Circumference of a circle



- 1 a 50.27 cm b 25.13 cm c 31.42 cm d 43.98 cm
- 2 169.6 cm
- 3 119.4 cm
- 4 2.23 cm
-

Arc length

- 5 a 12.6 units b 1.8 units c 6.2 m d 35.4 units
 e 62.8 cm
- 6 $r = 19.76$ cm
- 7 1.29 m
- 8 25 408.5 km
-

Area of circle

- 9 a 201.1 cm² b 78.5 cm² c 153.9 cm² d 113.1 cm²
- 10 a $r = 10.59$ cm b 66.5 cm
-

Sectors

- 11 a 18.9 cm b 18.8 cm

12

a

i 473.89 mm

ii 11 744.12 mm²



b

i 218.75 cm

ii 2683.95 cm²

c

i 92.40 m

ii 497.59 m²

d

i 108.91 mm

ii 730.62 mm²

13

a $P = 257.73 \text{ m}$

b $A = 3112.56 \text{ m}^2$

14

a $P = 586.78 \text{ cm}$

b $A = 15\,925.02 \text{ cm}^2$

Segment

15 $6\pi - 9\sqrt{3}$ square units

16 $16\sqrt{3} + \frac{160\pi}{3}$ square units

17

a $\theta = 30^\circ$

b $108\pi - 324 \text{ cm}^2$

c $1188\pi + 324 \text{ cm}^2$

d $11\pi + 3 : \pi - 3$

18

a 73.7°

b 6.4 cm

c 4.1 cm^2

19 50.7 cm^2

20 $\frac{13357\pi}{360} - \frac{361}{2} \sin 37^\circ \text{ mm}^2$

21

a $\theta = \frac{240^\circ}{\pi}$

b 14.6 cm^2

Applications

22

a $\theta = 60^\circ$

b $10\pi + 3\sqrt{3}$



23

a $x = 6$

b $\theta = 62^\circ$

c 9.8 cm^2

24

a $r = 12 \text{ cm}$

b $4\pi \text{ cm}$

25 22.1 laps

26

a 242.9 cm^2

b 242.9 cm^2

c 113.1 cm^2

d 146.1 cm^2

e 261.8 cm^2

f 40.8 cm^2

g 83.9 cm^2

h 176.5 units^2

i 19.6 cm^2

j 226.2 cm^2

27

a 60.68 cm

b 43.98 cm

c 42.85 m

d 35.00 cm

e 80.40 cm

f 31.42 cm

28

a 157.8 cm^2

b 9.4 cm^2

29

a 37.4 cm

b 35.7 mm

30 7.2 m

31

a 71 mm^2

b $\$284$

32 14.1 cm^2

33

a $r = 40.2 \text{ cm}^2$

b $r = 4.9 \text{ cm}$